Agricultural Water Quality Program

COLORADO STATE UNIVERSITY

Unveiling Biases in Water Sampling

A Bayesian Framework for Precision in Edge-of-Field Monitoring

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Introduction

Edge-of-field (EoF) water sampling is critical for tracking non-point source pollution.

Water is collected in many ways, often in the same location(s) due to limited resources

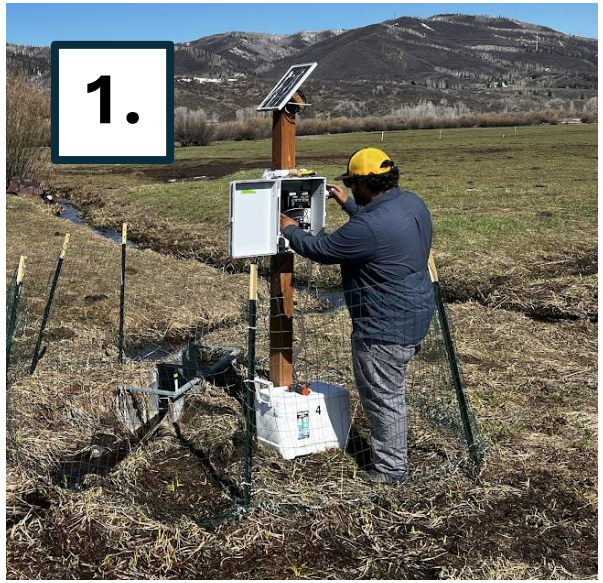
Despite widespread use, the accuracy and biases of common sampling methods are underexplored.

Biases can lead to misinterpretation of water quality data, affecting environmental assessments

EoF Installation at an ag field in Fort Collins, CO.
Cost: ~$15,000 for automated setup

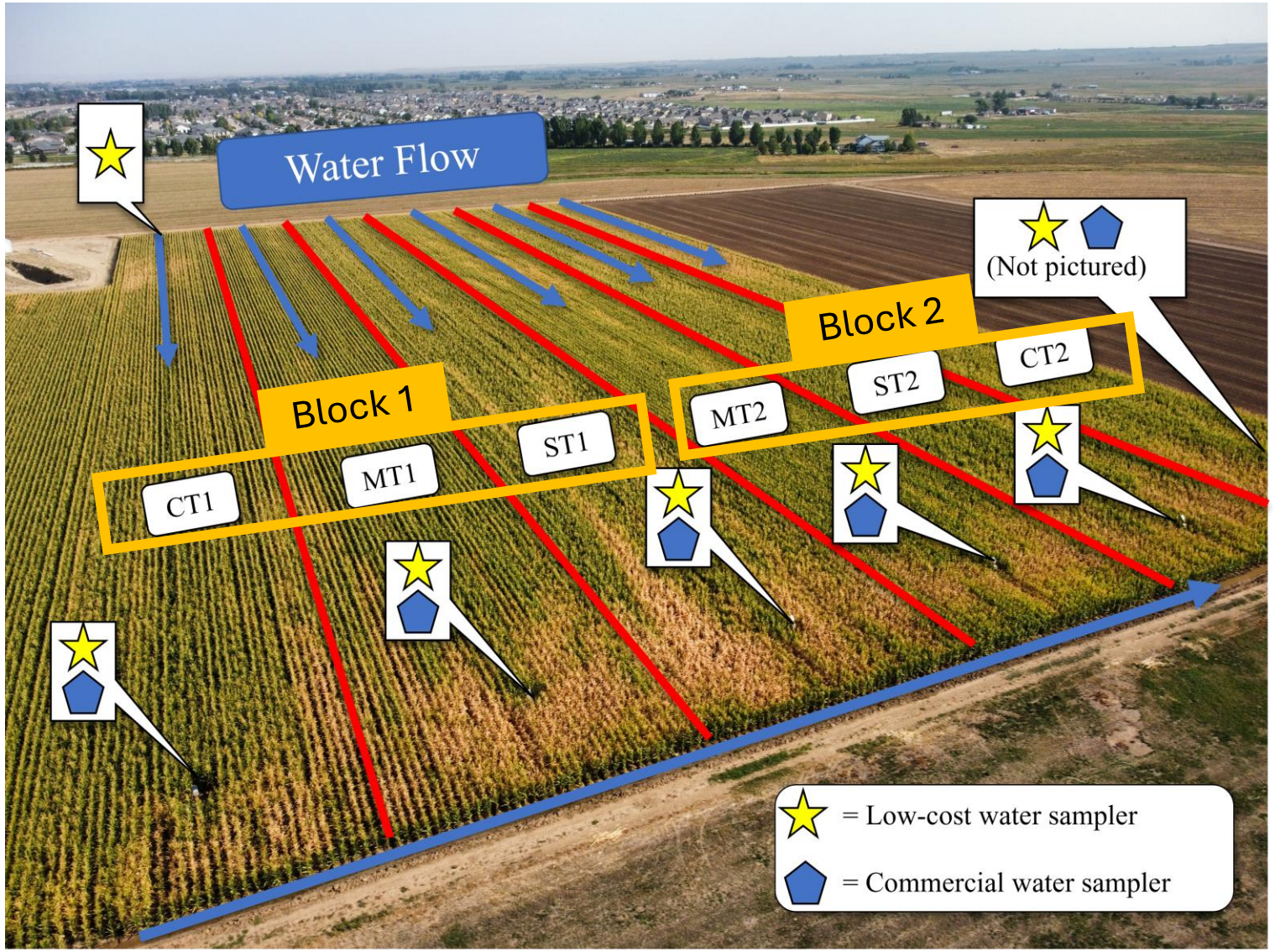
Objective

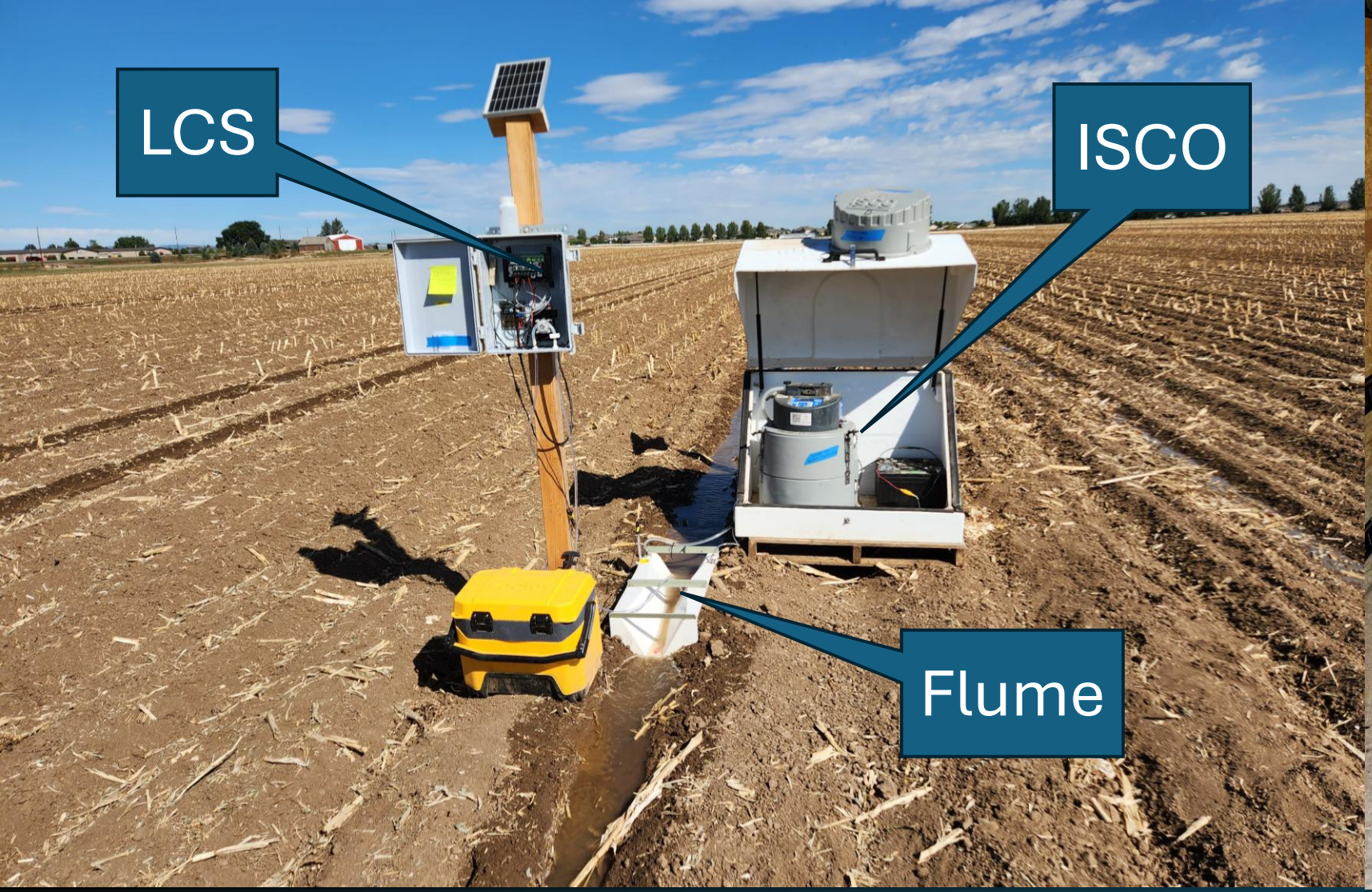
- Compare 4 different water sampling methods for 9 different water quality parameters
  1. **LCS:** low-cost automated sampler developed at CSU sampling hourly into a composite bottle
  2. **ISCO:** commercial automated sampler sampling hourly into a composite bottle
  3. **GBH:** water collected by hand each hour and put into a composite bottle
  4. **GB:** water collected by hand only at the “first flush”, the first hour into an event, and the last hour of an event
- **Goal:** to see if sampler method impacts WQ concentrations for all analytes <u>using a single statistical model</u>



Study Site and Experiment Layout

- Fort Collins, CO, USA
- 2023-2024
- Conservation Tillage Study started in 2011
  - CT = Full tillage
  - ST = Less tillage
  - MT = Least tillage
- Replicated management blocks
- Each plot is about 1 hectare with a 300m irrigation run length
- Surface irrigated with gated pipe for 7-10 hours at a time





Sampler Comparison Setup Up Close

Data Characteristics

- Events: 9 Irrigations, 2 Storms
- For each event:
  - Each sampling method collected a **primary and duplicate sample**
  - Water was taken to the lab, preserved, and tested for 9 analytes (see table)
- 454 unique water samples collected in total
  - 9 irrigations * 6 plots * 4 methods * 2 samples = 432
  - Storm 1 = 6 plots * 1 method * 2 samples = 18
  - Storm 2 = 2 plots * 1 method * 2 samples = 4
- Only LCS collected the storms b/c the ISCOs had no modem, and personnel could not be present to sample manually ← missing data problem (191 points)

Analyte	Method
Nitrogen, Nitrate (As N)	EPA 300.0
Nitrogen, Nitrite (As N)	EPA 300.0
Nitrogen, Total Kjeldahl	M4500 NH3 D - TKN_W 4500NH3 D
Phosphorus, Total (As P)	SM4500P E - P
Phosphorus, Total Orthophosphate (as P)	SM4500P E - P-Ortho
Suspended Solids (Residue, Non-Filterable)	EPA 160.2
Total Dissolved Solids	EPA 160.1
Specific Electrical Conductivity	EPA 120.1
pH	EPA 150.1

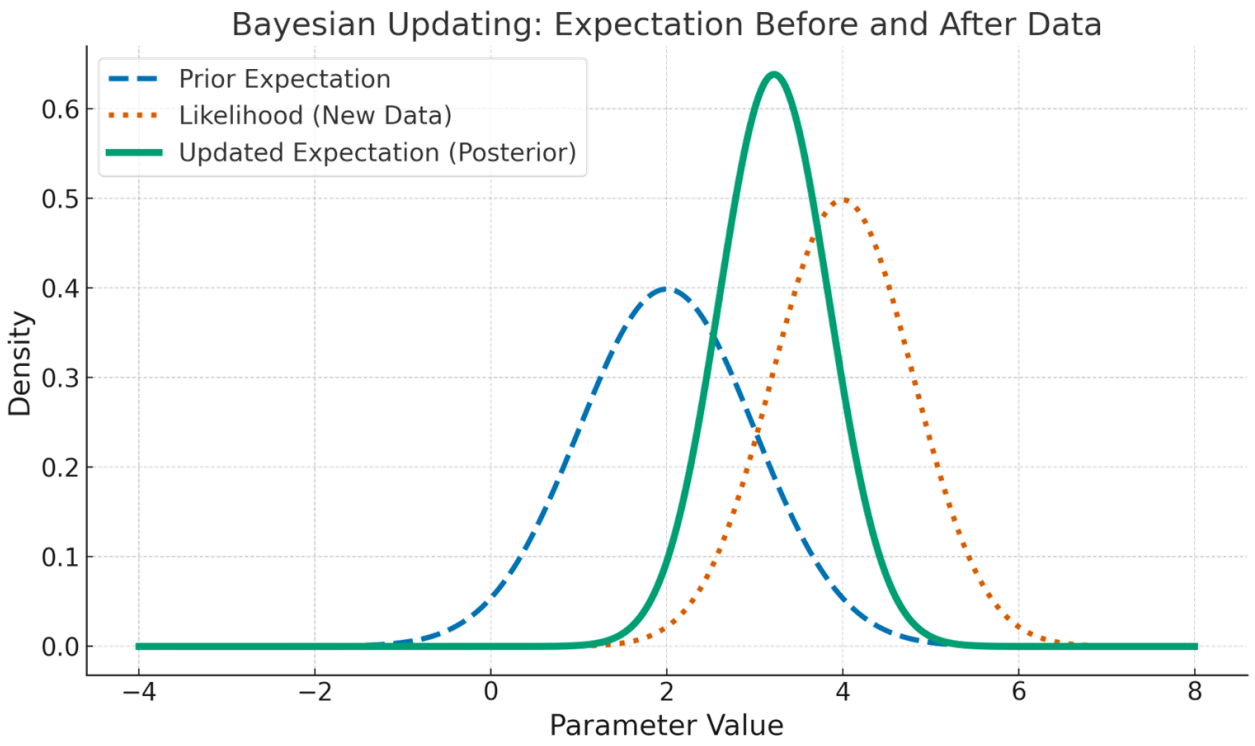
Statistical Approach: Bayesian

Pros

- Result is probability distribution(s)
- Handles uncertainty inherently
- Can estimate and use missing data during model calibration
- Minimum sample size is 1

Cons

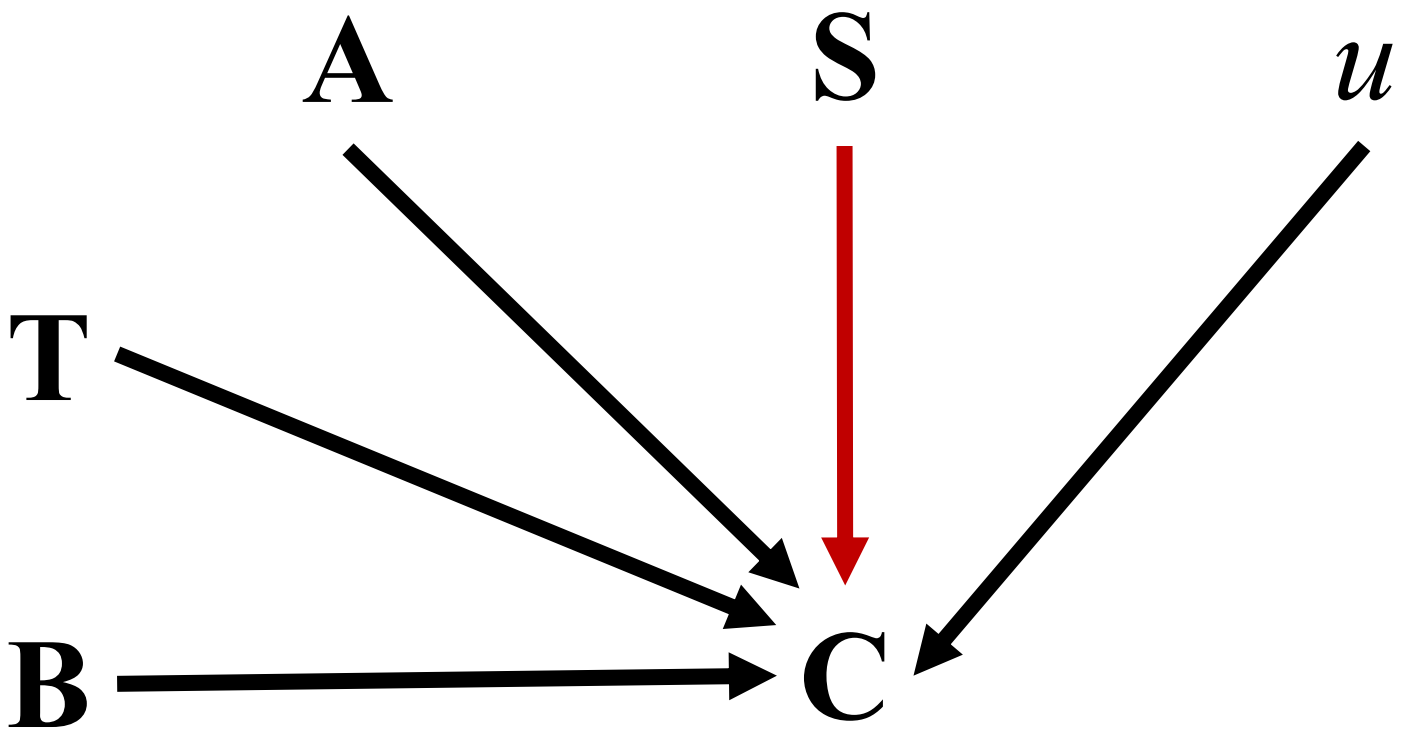
- Computationally intensive
- Sensitive to choice of priors and model form, especially with limited data, emphasizing importance of expertise



What Scientists Want: Scientists often want to know how likely their hypothesis is to be true given the data, i.e., $P(H_0|data)$, which is a Bayesian concept (posterior probability).

Frequentist Limitation: The p-value tells you the probability of observing your data (or something more extreme) if the null hypothesis is already true i.e., $P(data|H_0)$. However, many mistakenly interpret this as the probability that the null hypothesis is true given the data i.e., $P(H_0|data)$.

Establishing Assumptions - Causal Model



Arrows = directional causes

There is also other uncertainty in our observations (e.g., lab measurement error).

The sampler method (S), analyte of interest (A), tillage treatment (T), and experimental block (B) all influence the observed concentration (C). Analyte impacts may vary depending on S, T, and B, so we must control for them in our model. We want to understand the direct effect of S on C

Statistical Model: Generalized Linear Mixed Model (GLMM)

$C_{Obs,i} \sim Normal(\mu_i, \sigma)$

Prior for expected z-score concentration, *i*

$\mu_i = \alpha_{A[i]} + \beta_{A[i],S[i]} + \gamma_{A[i],B[i]} + \Delta_{A[i],T[i]}$

Mean + Analyte-Sampler + Analyte-Block + Analyte-Treatment

$\sigma \sim Exponential(1)$

Unobserved uncertainty prior

$\alpha_A \sim Normal(0,1)$

Prior for each 'true' analyte mean conc. After standardizing each to a z-score

$\beta_{A,S} \sim MVN([0,0,0,0], \mathbf{R}_S, \mathbf{S}_S)$

Prior for covarying analyte-sampler effects
A vector of 4 parameters for each sampler

$\gamma_{A,B} \sim MVN([0,0], \mathbf{R}_B, \mathbf{S}_B)$

Prior for covarying analyte-block effects
A vector of 2 parameters for each block

$\Delta_{A,T} \sim MVN([0,0,0], \mathbf{R}_T, \mathbf{S}_T)$

Prior for covarying analyte-treatment effects
A vector of 3 parameters for each treatment

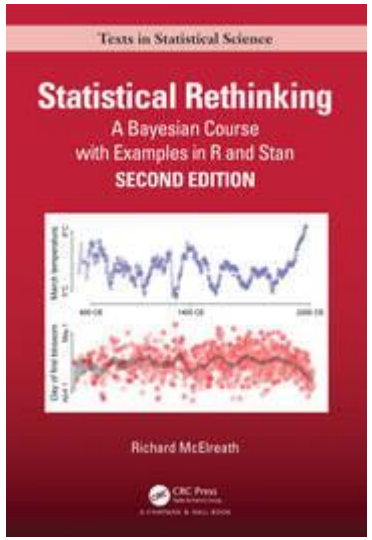
$\mathbf{S}_S, \mathbf{S}_B, \mathbf{S}_T \sim Exponential(1)$

All std. dev.'s get exp. (1) prior
1 for each sampler, block, and treatment

$\mathbf{R}_S, \mathbf{R}_B, \mathbf{R}_T \sim LKJcorr(2)$

Correlation matrix prior (a prior expectation of covariance) for each MVN distribution

Translating model to R code



Rethinking package

Stan Software

```r
# Non-centered version
m1.6_nc <- ulam(
  alist(
    # Observation model
    C_obs ~ dnorm(mu, sigma),

    # Mean structure with correlations between analytes, samplers, and blocks
    mu <- alpha[A] + beta[A, S] + gamma[A, B] + delta[A, TRT],

    # Analyte-specific (9 levels) intercepts (non-centered parameterization)
    transpars> vector[K_A]:alpha <<- mu_alpha + z_alpha * sigma_alpha,
    vector[K_A]:z_alpha ~ normal(0, 1),  # Non-centered scaling
    mu_alpha ~ normal(0, 1),  # Prior for mean intercepts
    sigma_alpha ~ exponential(1),  # Prior for intercept variation

    # Analyte-specific (9 levels) sampler (4 levels) effects (non-centered parameterization)
    transpars> matrix[K_A, K_S]:beta <<- mu_beta + z_beta * diag_pre_multiply(sigma_beta, L_beta),
    transpars> matrix[K_A, K_S]:mu_beta <- rep_matrix(0, K_A, K_S),  # Center at zero
    matrix[K_A, K_S]:z_beta ~ normal(0, 1),  # Standardized effects
    transpars> cholesky_factor_corr[K_S]:L_beta ~ lkj_corr_cholesky(2),  # LKJ prior
    vector[K_S]:sigma_beta ~ exponential(1),  # Prior for scaling sampler effects

    # Analyte-specific (9 levels) block (2 levels) effects (non-centered parameterization)
    transpars> matrix[K_A, K_B]:gamma <<- mu_gamma + z_gamma * diag_pre_multiply(sigma_gamma, L_gamma),
    transpars> matrix[K_A, K_B]:mu_gamma <- rep_matrix(0, K_A, K_B),  # Center at zero
    matrix[K_A, K_B]:z_gamma ~ normal(0, 1),  # Standardized effects
    transpars> cholesky_factor_corr[K_B]:L_gamma ~ lkj_corr_cholesky(2),  # LKJ prior
    vector[K_B]:sigma_gamma ~ exponential(1),  # Prior for scaling block effects

    # Analyte-specific (9 levels) treatment (3 levels) effects (non-centered parameterization)
    transpars> matrix[K_A, K_T]:delta <<- mu_delta + z_delta * diag_pre_multiply(sigma_delta, L_delta),
    transpars> matrix[K_A, K_T]:mu_delta <- rep_matrix(0, K_A, K_T),  # Center at zero
    matrix[K_A, K_T]:z_delta ~ normal(0, 1),  # Standardized effects
    transpars> cholesky_factor_corr[K_T]:L_delta ~ lkj_corr_cholesky(2),  # LKJ prior
    vector[K_T]:sigma_delta ~ exponential(1),  # Prior for scaling treatment effects

    # Prior for measurement error
    sigma ~ exponential(1)
  ),
  data = data1.6,
  chains = 4,
  cores = 12,
  iter = 4000,              # Total iterations (increase this)
  warmup = 1000             # Number of warmup iterations (optional, default is iter/2)
)
```

Data simulation

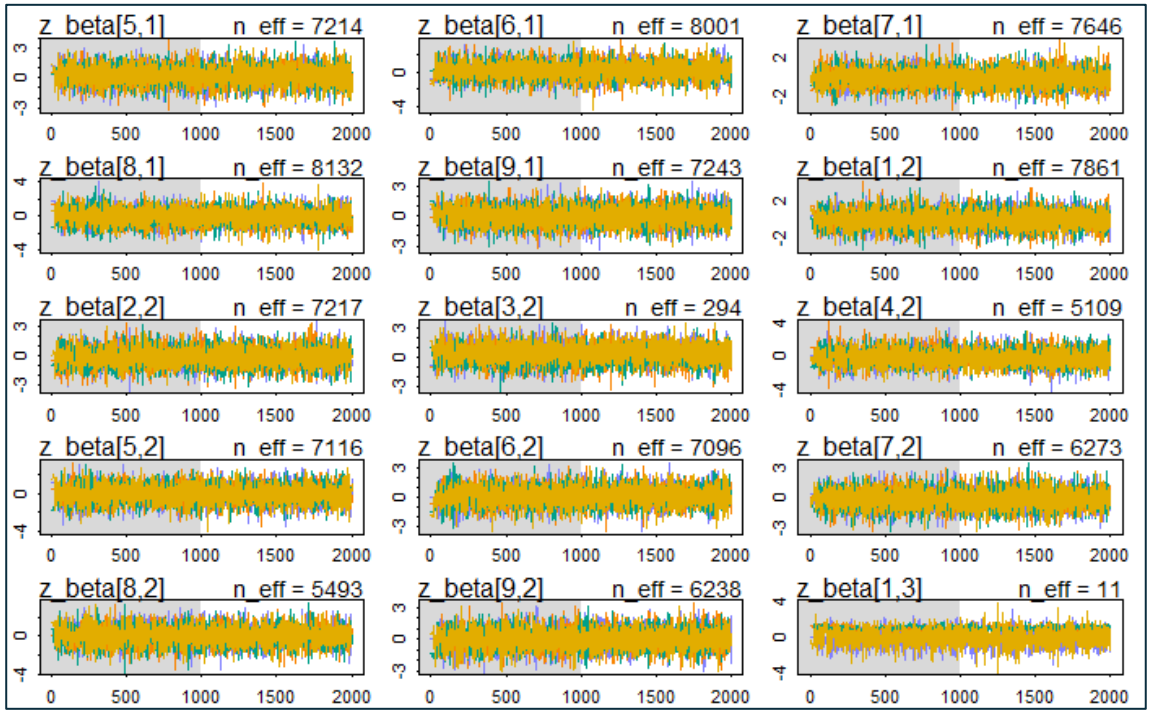
```{r}
simulate_model_1.3 <- function(n_per_type = 100, noise_sd = 0.2, seed = 45) {
  library(dplyr)

  # Define sampler types and their true intercepts
  sampler_types <- c("LCS", "ISCO", "GB", "GBH")
  true_intercepts <- c(LCS = 0.7, ISCO = 0.3, GB = 1.2, GBH = 1.0)

  # Define analytes' base means and standard deviations
  base_means <- c(NO3 = 8, NO2 = 0.1, TKN = 5, pH = 7, TP = 0.8, OP = 0.3, EC = 0.15, TSS = 1000, TDS =
  base_sd <- c(NO3 = 1, NO2 = 0.01, TKN = 1, pH = 0.1, TP = 0.1, OP = 0.05, EC = 0.01, TSS = 200, TDS =

  # Define block-specific effects
  block_effects <- c(Block1 = 0, Block2 = 0.5)

  # Set seed for reproducibility
  set.seed(seed)

  # Create all combinations of analytes, sampler types, and blocks
  sim_data <- expand.grid(
    analyte_abbr = seq_along(base_means), # Numeric index for analytes
    S = sampler_types,
    block = names(block_effects)
```

Important Step: Data was simulated with known influences for each parameter and used to test the model for accuracy

Calibrating the model on Real Data

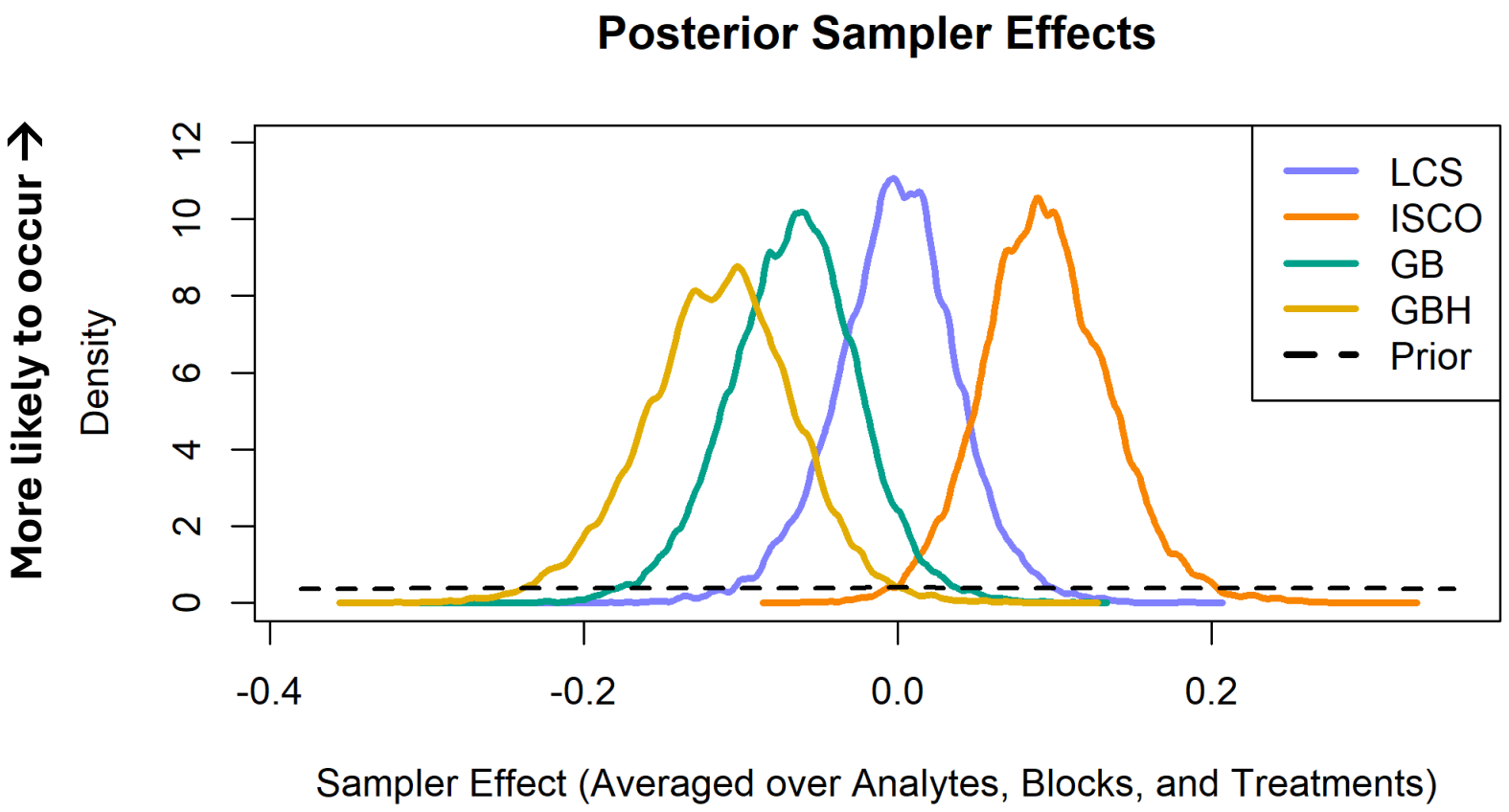
- Hamiltonian Monte Carlo (HMC) sampling explores the data and "**converges**" on parameter values
  - We used 4 "chains", sampling 4000 times each
- `rhat` = Gelman-Rubin statistic
  - $\hat{R} \approx 1.0$: Indicates convergence; the chains are sampling from the same distribution, and the posterior estimates are likely reliable.
- `ess_bulk` = effective sampling size
  - Larger = better; can be larger than number of samples when converging well and efficiently
- Bayes lets us impute missing data, allowing for more efficient model calibration, also improving accuracy

"Fuzzy Caterpillars" = good exploration and convergence




```
> precis(updated_model_real, depth = 2)
272 matrix parameters hidden. Use depth=3 to show them
                  mean   sd  5.5% 94.5% rhat ess_bulk
z_alpha[1]        0.04 0.93 -1.43  1.53    1 22760.80
z_alpha[2]       -0.05 0.94 -1.53  1.46    1 20385.05
z_alpha[3]        0.00 0.95 -1.51  1.50    1 17045.66
z_alpha[4]        0.00 0.94 -1.49  1.52    1 20818.54
z_alpha[5]        0.03 0.94 -1.48  1.53    1 18928.46
z_alpha[6]       -0.05 0.94 -1.56  1.45    1 20703.08
z_alpha[7]       -0.07 0.95 -1.58  1.46    1 20611.47
z_alpha[8]        0.02 0.93 -1.47  1.49    1 22234.57
```

Interpreting Results – Effect of sampler overall



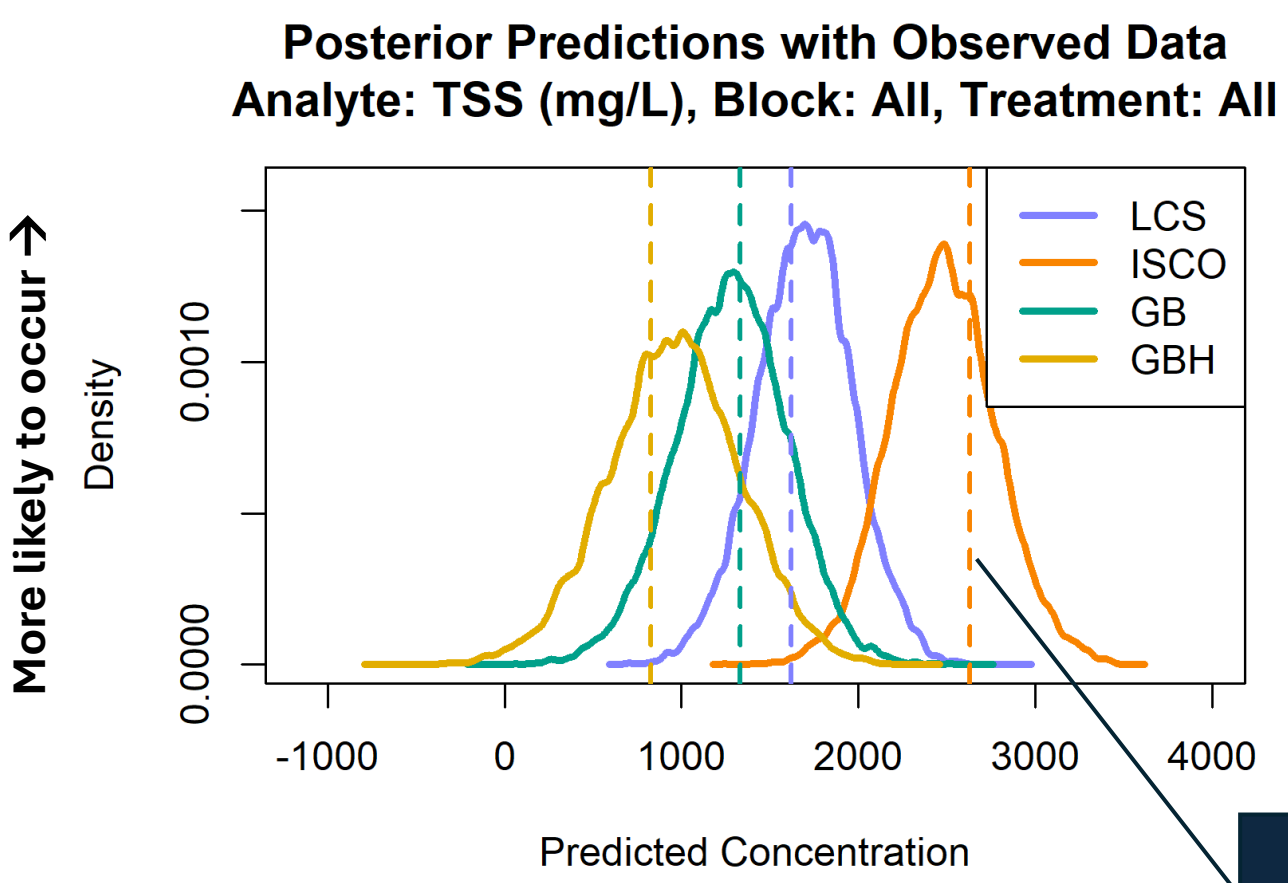
What this means

- LCS appears to be the least biased (i.e., centered over 0)
- ISCO is positively biased
- GB and GBH are negatively biased
- Prior is shown to ensure that sampling resulted in meaningful change

X-axis: 0 = no effect on measurement, >0 = positive bias, <0 = negative bias

Averaged over all analytes

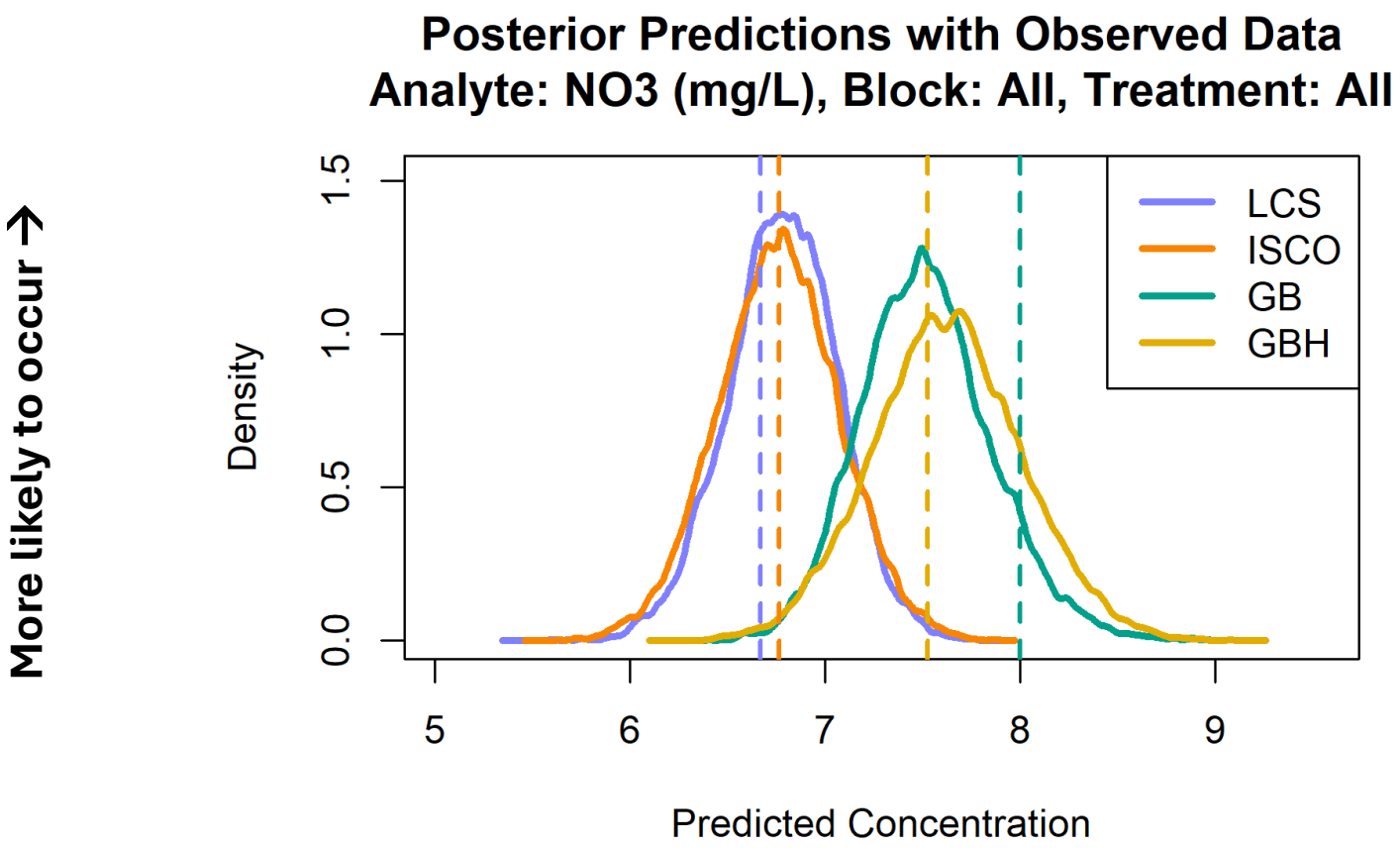
Generative model predictions – results for mortals



Observed averages are dotted vertical lines

- Total Suspended Solids (TSS)
  - Averaged over all irrigations, blocks, and treatments
- Notice how the ISCO is much more likely to measure higher values
- GBH has lowest measurement
  - Sediment might get stuck in flume
  - Might be less disturbing of settled media relative to ISCO and LCS
  - GB might have positive TSS bias due to 1/3 of volume being First Flush.

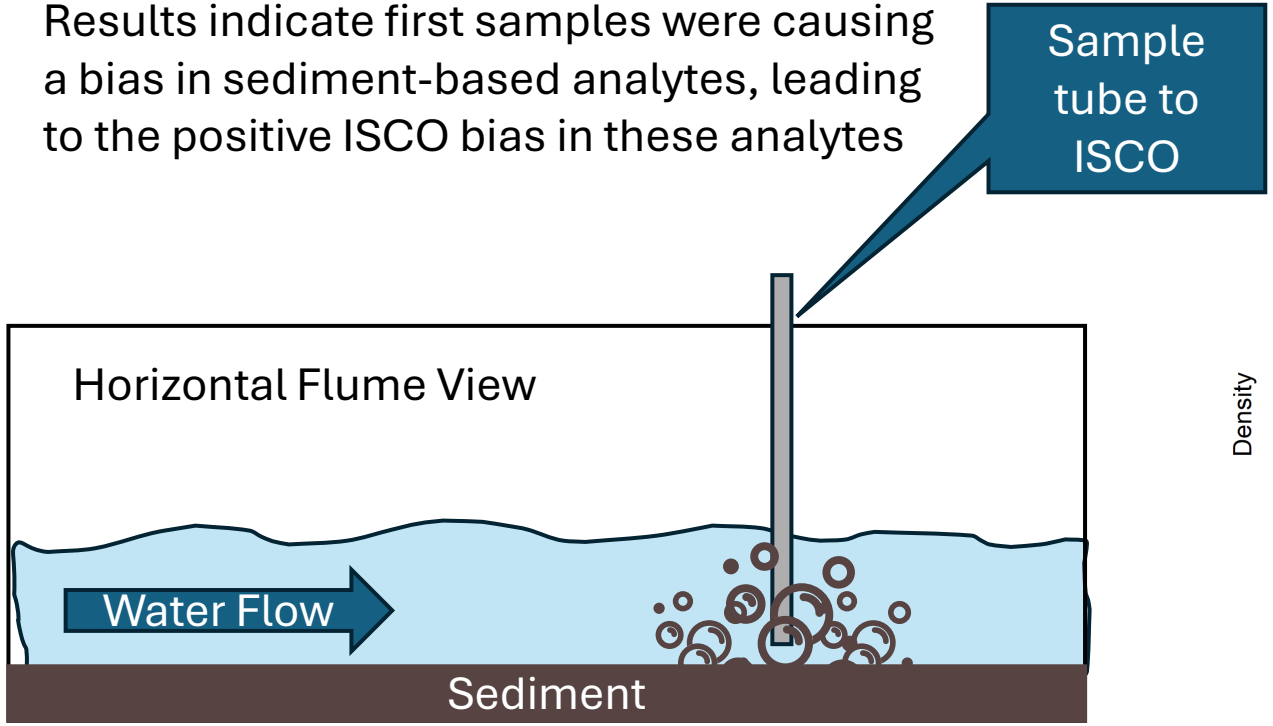
Now NO3 to compare



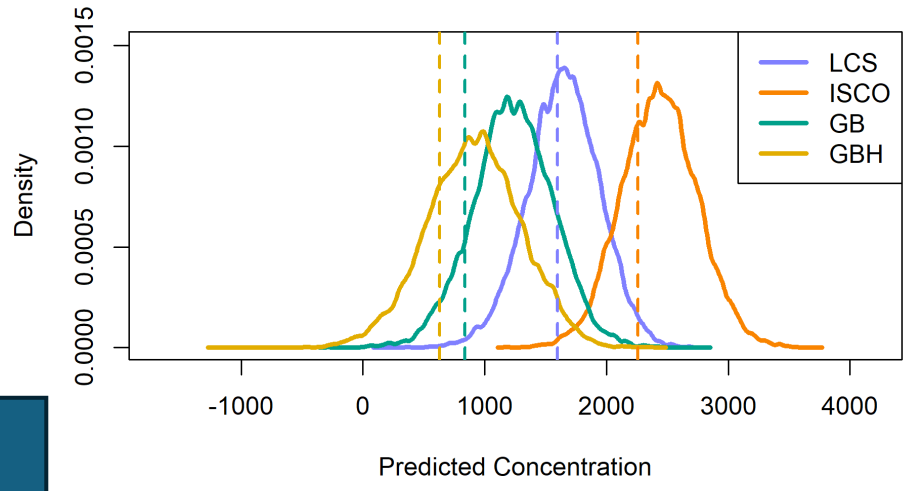
- ISCO and LCS are in good agreement at around 6.25 ppm on average
- GB and GBH are in good agreement at the 7-8 ppm range
- Soluble analyte, likely led to better agreement because water disturbance is less of a factor

Plume Theory

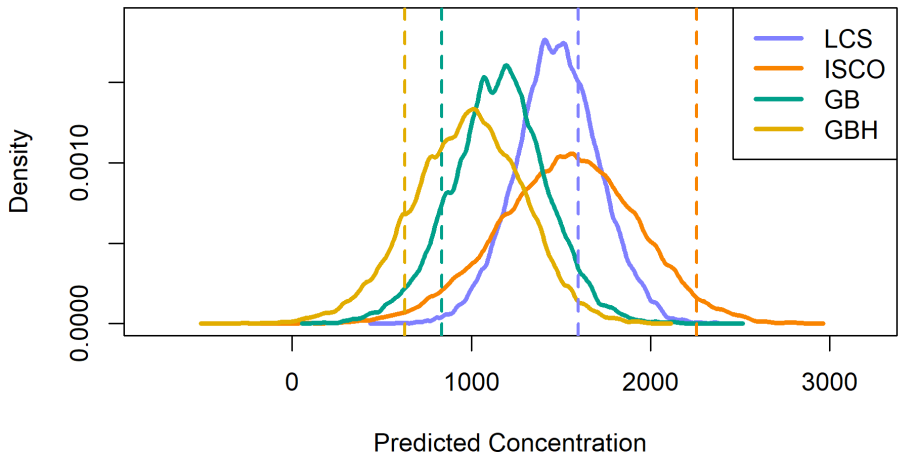
- ISCO's purge air down the flume sample tube prior to sampling, creating a "plume" of sediment that gets sampled during the first of the two duplicate water samples
- We removed the first water sample taken by ISCOs and reran the model
- Results indicate first samples were causing a bias in sediment-based analytes, leading to the positive ISCO bias in these analytes

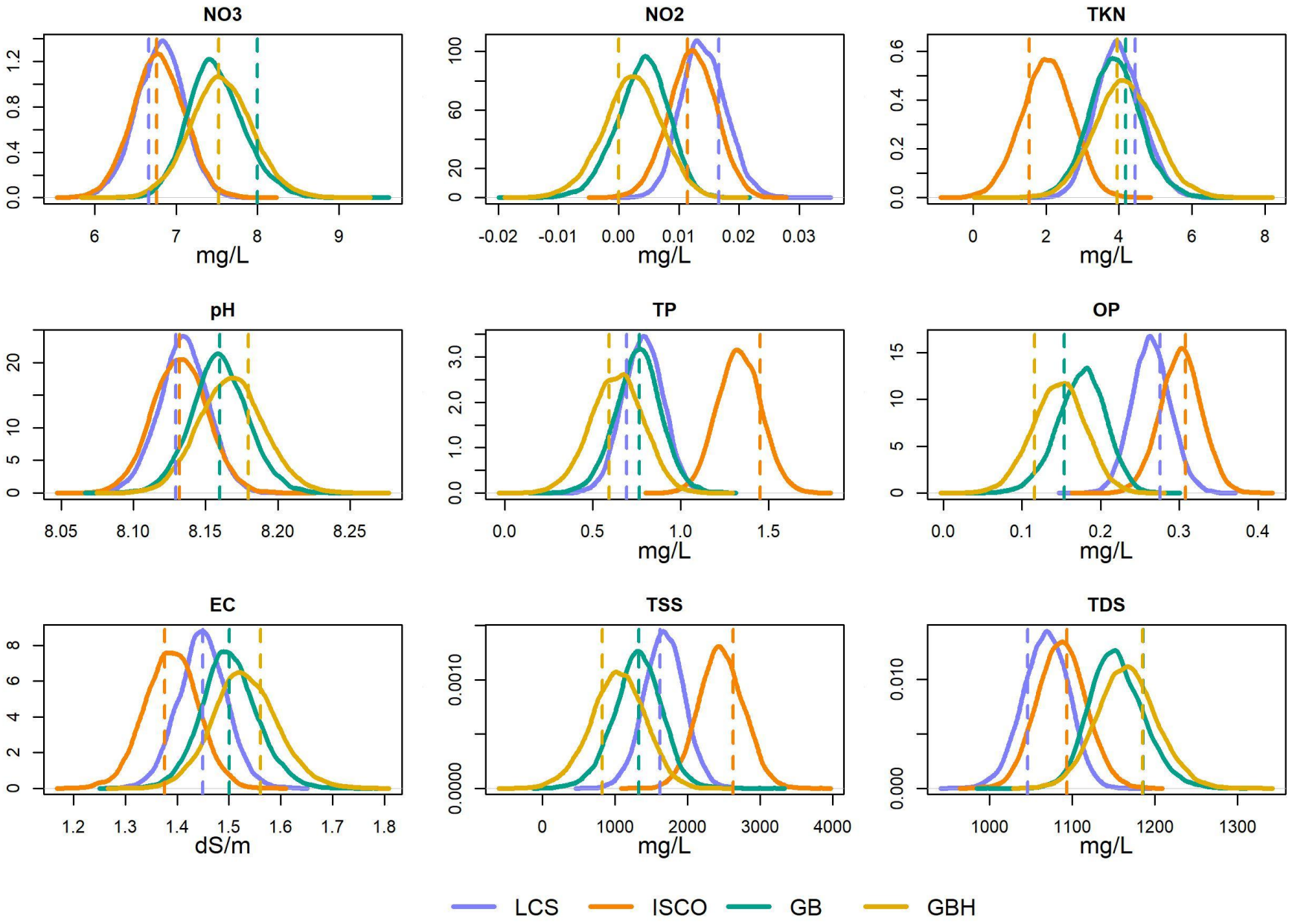


TSS (mg/L) Original Likelihood Distributions



TSS (mg/L) First Samples Removed





All Analytes

- Model appears to be doing well at capturing observed results, supporting it's use for our causal inference
- Sampler effects can vary greatly by analyte, but were captured well by our model

Conclusions and Future Work

Sampling method matters, but it depends a lot on what's being tested

- TSS (Insoluble) = ISCO biased high
- NO3 (Soluble) = ISCO not very biased to other methods
- Unsure of exact mechanisms on this; plume theory is leading explanation

The Bayesian approach allowed us to tease out these results

- Modeling covariance improved model fit and interpretation
- We imputed missing data instead of ignoring it, improving model efficiency
- It created a generative model that can be used for forecasting with probabilistic distributions instead of point estimates (like machine learning but with a causal result and much less data needed to calibrate)

Future work

- Models can always be more complicated (e.g., adding temporal nature of irrigation events)
- Calculate loads using flow data
- Incorporate 13-yr dataset at this site using all tested methods of sampling
- Ultimately find the impact of tillage management on runoff water quality to guide farmers to better practices

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Thank You